Human audit sheet: rules and answer keys

Please read each item and tick one box. About 20 minutes. All rules are in English; the words are invented, so only the logic matters.

The two rule forms mean:

- *Every A is a B*: anything that is an A is also a B.
- *Every A that is also a B is a C*: anything that is both an A and a B is also a C.

Part 1. Is the fact entailed?

The key says the last line of each derivation is entailed. Check that every step follows from the line before and the rules shown.

Item 1 (problem P4). Start: *Nell is a nembaxix.*

- Every nembaxix is a xezlager. → Nell is a xezlager.
- Every xezlager is a norzejow. → Nell is a norzejow.
- Every norzejow is a zujnixim. → Nell is a zujnixim.
- Every zujnixim is a koxtotad. → Nell is a koxtotad.
- Every koxtotad is a firkigeg. → Nell is a firkigeg.
- Every firkigeg is a serzadin. → Nell is a serzadin.
- Every serzadin is a bixtuwoz. → Nell is a bixtuwoz.
- Every bixtuwoz is a gajnawut. → Nell is a gajnawut.
- Every gajnawut is a daskovaf. → Nell is a daskovaf.
- Every daskovaf is a zukredul. → Nell is a zukredul.
- Every zukredul is a telvolid. → Nell is a telvolid.
- Every telvolid is a kirkinam. → Nell is a kirkinam.
- Every kirkinam is a xonkunim. → Nell is a xonkunim.
- Every xonkunim is a bozlazin. → Nell is a bozlazin.
- Every bozlazin is a pimbebev. → Nell is a pimbebev.
- Every pimbebev is a saskotix. → Nell is a saskotix.
- Every saskotix is a voskazop. → Nell is a voskazop.
- Every voskazop is a renkixak. → Nell is a renkixak.
- Every renkixak is a vikropaz. → Nell is a vikropaz.
- Every vikropaz is a zojnovus. → Nell is a zojnovus.

Key says: **Nell is a zojnovus** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 2 (problem P3). Start: *Ines is a vuxtalev.*

- Every vuxtalev is a wutlibis. → Ines is a wutlibis.
- Every wutlibis is a wokremus. → Ines is a wokremus.
- Every wokremus is a xetlizob. → Ines is a xetlizob.

- Every xetlizob is a gigniver. → Ines is a gigniver.

Key says: **Ines is a gigniver** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 3 (problem P1). Start: *Orla is a sordijum.*

- Every sordijum is a lejnavob. → Orla is a lejnavob.
- Every lejnavob is a lozlesow. → Orla is a lozlesow.
- Every lozlesow is a xezmuzuv. → Orla is a xezmuzuv.
- Every xezmuzuv is a wakrexok. → Orla is a wakrexok.
- Every wakrexok is a juxtujaj. → Orla is a juxtujaj.
- Every juxtujaj is a nezmepag. → Orla is a nezmepag.
- Every nezmepag is a voltozor. → Orla is a voltozor.
- Every voltozor is a xuvkonuw. → Orla is a xuvkonuw.
- Every xuvkonuw is a xojnijet. → Orla is a xojnijet.
- Every xojnijet is a wodrined. → Orla is a wodrined.
- Every wodrined is a bovkuwup. → Orla is a bovkuwup.
- Every bovkuwup is a deskesop. → Orla is a deskesop.
- Every deskesop is a javkokap. → Orla is a javkokap.
- Every javkokap is a dinkemef. → Orla is a dinkemef.
- Every dinkemef is a jiltezed. → Orla is a jiltezed.
- Every jiltezed is a mozmesuf. → Orla is a mozmesuf.
- Every mozmesuf is a zazmizen. → Orla is a zazmizen.
- Every zazmizen is a lovkusez. → Orla is a lovkusez.
- Every lovkusez is a salvagig. → Orla is a salvagig.

Key says: **Orla is a salvagig** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 4 (problem P8). Start: *Tobin is a buvkilib.*

- Every buvkilib is a zekravuw. → Tobin is a zekravuw.
- Every zekravuw is a durdowef. → Tobin is a durdowef.
- Every durdowef is a pekrawis. → Tobin is a pekrawis.
- Every buvkilib is a wozlovem. → Tobin is a wozlovem.
- Every wozlovem is a buskotot. → Tobin is a buskotot.
- Every buskotot is a peskabet. → Tobin is a peskabet.
- Every peskabet is a norkijok. → Tobin is a norkijok.
- Every norkijok is a xutlomag. → Tobin is a xutlomag.
- Every xutlomag is a vixtawef. → Tobin is a vixtawef.
- Every pekrawis that is also a vixtawef is a delvuzov. → Tobin is a delvuzov.
- Every buvkilib is a zekravuw. → Tobin is a zekravuw.
- Every zekravuw is a durdowef. → Tobin is a durdowef.
- Every durdowef is a pekrawis. → Tobin is a pekrawis.
- Every pekrawis is a girzomew. → Tobin is a girzomew.

- Every buvkilib is a zozlebos. → Tobin is a zozlebos.
- Every zozlebos is a pinkubun. → Tobin is a pinkubun.
- Every pinkubun is a koskuxem. → Tobin is a koskuxem.
- Every koskuxem is a mozmixix. → Tobin is a mozmixix.
- Every mozmixix is a waxtovis. → Tobin is a waxtovis.
- Every waxtovis is a vambesev. → Tobin is a vambesev.
- Every girzomew that is also a vambesev is a jezmijaw. → Tobin is a jezmijaw.
- Every delvuzov that is also a jezmijaw is a mirkenub. → Tobin is a mirkenub.
- Every mirkenub is a zorzereg. → Tobin is a zorzereg.

Key says: **Tobin is a zorzereg** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 5 (problem P8). Start: *Tobin is a buvkilib.*

- Every buvkilib is a furdejik. → Tobin is a furdejik.
- Every furdejik is a monzelez. → Tobin is a monzelez.
- Every monzelez is a zoltukuf. → Tobin is a zoltukuf.
- Every zoltukuf is a zivkukat. → Tobin is a zivkukat.
- Every zivkukat is a lilvogat. → Tobin is a lilvogat.
- Every lilvogat is a sunkozog. → Tobin is a sunkozog.
- Every sunkozog is a filtuken. → Tobin is a filtuken.
- Every filtuken is a dedriwun. → Tobin is a dedriwun.
- Every dedriwun is a kembowox. → Tobin is a kembowox.
- Every buvkilib is a jetlotis. → Tobin is a jetlotis.
- Every jetlotis is a vivkilem. → Tobin is a vivkilem.
- Every vivkilem is a dekrajez. → Tobin is a dekrajez.
- Every dekrajez is a jezmabej. → Tobin is a jezmabej.
- Every kembowox that is also a jezmabej is a jatlimip. → Tobin is a jatlimip.
- Every buvkilib is a zozlebos. → Tobin is a zozlebos.
- Every zozlebos is a pinkubun. → Tobin is a pinkubun.
- Every pinkubun is a koskuxem. → Tobin is a koskuxem.
- Every koskuxem is a mozmixix. → Tobin is a mozmixix.
- Every buvkilib is a jiskesew. → Tobin is a jiskesew.
- Every jiskesew is a rejnidum. → Tobin is a rejnidum.
- Every rejnidum is a miltewiz. → Tobin is a miltewiz.
- Every miltewiz is a wurzaxux. → Tobin is a wurzaxux.
- Every wurzaxux is a zunzafev. → Tobin is a zunzafev.
- Every zunzafev is a danzaniz. → Tobin is a danzaniz.
- Every danzaniz is a zoxtixaj. → Tobin is a zoxtixaj.
- Every zoxtixaj is a textaxud. → Tobin is a textaxud.
- Every textaxud is a rekrikuk. → Tobin is a rekrikuk.
- Every rekrikuk is a geskujet. → Tobin is a geskujet.
- Every mozmixix that is also a geskujet is a wuzmasom. → Tobin is a wuzmasom.

- Every jatlimip that is also a wuzmasom is a zoltediz. → Tobin is a zoltediz.
- Every buvkilib is a jetlotis. → Tobin is a jetlotis.
- Every jetlotis is a vivkilem. → Tobin is a vivkilem.
- Every vivkilem is a dekrajez. → Tobin is a dekrajez.
- Every dekrajez is a jezmabej. → Tobin is a jezmabej.
- Every jezmabej is a revkusod. → Tobin is a revkusod.
- Every revkusod is a takrigim. → Tobin is a takrigim.
- Every takrigim is a binkevaj. → Tobin is a binkevaj.
- Every binkevaj is a wizlirar. → Tobin is a wizlirar.
- Every wizlirar is a pezmemav. → Tobin is a pezmemav.
- Every pezmemav is a nijnutij. → Tobin is a nijnutij.
- Every nijnutij is a berzoran. → Tobin is a berzoran.
- Every berzoran is a porzulat. → Tobin is a porzulat.
- Every porzulat is a jedrefad. → Tobin is a jedrefad.
- Every jedrefad is a bozlusat. → Tobin is a bozlusat.
- Every bozlusat is a pakreruf. → Tobin is a pakreruf.
- Every zoltediz that is also a pakreruf is a noskemak. → Tobin is a noskemak.
- Every noskemak is a tikrezug. → Tobin is a tikrezug.

Key says: **Tobin is a tikrezug** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 6 (problem P2). Start: *Rex is a zerzofox.*

- Every zerzofox is a workanow. → Rex is a workanow.
- Every workanow is a luzmozid. → Rex is a luzmozid.
- Every luzmozid is a teskepoz. → Rex is a teskepoz.
- Every teskepoz is a faskonuf. → Rex is a faskonuf.
- Every faskonuf is a jugnelum. → Rex is a jugnelum.
- Every jugnelum is a mejnoged. → Rex is a mejnoged.

Key says: **Rex is a mejnoged** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 7 (problem P4). Start: *Nell is a nembaxix.*

- Every nembaxix is a jurdufox. → Nell is a jurdufox.
- Every jurdufox is a dilvabit. → Nell is a dilvabit.
- Every dilvabit is a funzevip. → Nell is a funzevip.
- Every nembaxix is a tezmerat. → Nell is a tezmerat.
- Every tezmerat is a sanzataw. → Nell is a sanzataw.
- Every sanzataw is a wizlitaf. → Nell is a wizlitaf.
- Every wizlitaf is a mimbobex. → Nell is a mimbobex.
- Every mimbobex is a zarkabev. → Nell is a zarkabev.
- Every funzevip that is also a zarkabev is a lugnoxav. → Nell is a lugnoxav.
- Every nembaxix is a bazmakuw. → Nell is a bazmakuw.

- Every bazmakuw is a torkelek. → Nell is a torkelek.
- Every torkelek is a garkuteg. → Nell is a garkuteg.
- Every garkuteg is a tazmonog. → Nell is a tazmonog.
- Every tazmonog is a tezlubem. → Nell is a tezlubem.
- Every tezlubem is a jatlofet. → Nell is a jatlofet.
- Every jatlofet is a zazlunus. → Nell is a zazlunus.
- Every zazlunus is a fezmogag. → Nell is a fezmogag.
- Every fezmogag is a maskipup. → Nell is a maskipup.
- Every maskipup is a kumbomul. → Nell is a kumbomul.
- Every kumbomul is a lukrakux. → Nell is a lukrakux.
- Every lukrakux is a dumbovaj. → Nell is a dumbovaj.
- Every dumbovaj is a xezmuxed. → Nell is a xezmuxed.
- Every xezmuxed is a jultesam. → Nell is a jultesam.
- Every lugnoxav that is also a jultesam is a nerziruv. → Nell is a nerziruv.
- Every nembaxix is a xirdefad. → Nell is a xirdefad.
- Every xirdefad is a metlaxom. → Nell is a metlaxom.
- Every metlaxom is a jodragup. → Nell is a jodragup.
- Every jodragup is a vuskixad. → Nell is a vuskixad.
- Every vuskixad is a juskubos. → Nell is a juskubos.
- Every juskubos is a xordedag. → Nell is a xordedag.
- Every xordedag is a juzmizes. → Nell is a juzmizes.
- Every nembaxix is a giskerad. → Nell is a giskerad.
- Every giskerad is a vugnavef. → Nell is a vugnavef.
- Every vugnavef is a ruskurep. → Nell is a ruskurep.
- Every ruskurep is a xoxtexuj. → Nell is a xoxtexuj.
- Every xoxtexuj is a rejnoded. → Nell is a rejnoded.
- Every rejnoded is a jazlenok. → Nell is a jazlenok.
- Every juzmizes that is also a jazlenok is a ferkopis. → Nell is a ferkopis.
- Every nembaxix is a xezlager. → Nell is a xezlager.
- Every xezlager is a norzejow. → Nell is a norzejow.
- Every norzejow is a zujnixim. → Nell is a zujnixim.
- Every nembaxix is a tezmerat. → Nell is a tezmerat.
- Every tezmerat is a sanzataw. → Nell is a sanzataw.
- Every sanzataw is a wizlitaf. → Nell is a wizlitaf.
- Every wizlitaf is a mimbobex. → Nell is a mimbobex.
- Every mimbobex is a zarkabev. → Nell is a zarkabev.
- Every zarkabev is a rukruleb. → Nell is a rukruleb.
- Every rukruleb is a pekruxow. → Nell is a pekruxow.
- Every pekruxow is a sarzapij. → Nell is a sarzapij.
- Every sarzapij is a molvezob. → Nell is a molvezob.
- Every molvezob is a tanzarun. → Nell is a tanzarun.

- Every zujnixim that is also a tanzarun is a nurzosiw. → Nell is a nurzosiw.
- Every ferkopis that is also a nurzosiw is a jembizin. → Nell is a jembizin.
- Every nerziruv that is also a jembizin is a xunzuzoz. → Nell is a xunzuzoz.

Key says: **Nell is a xunzuzoz** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 8 (problem P5). Start: *Ines is a goskomab.*

- Every goskomab is a xombokaz. → Ines is a xombokaz.
- Every xombokaz is a vivkabaz. → Ines is a vivkabaz.
- Every vivkabaz is a rizmizur. → Ines is a rizmizur.
- Every rizmizur is a wavkebif. → Ines is a wavkebif.
- Every wavkebif is a rizlabed. → Ines is a rizlabed.
- Every rizlabed is a solvilaw. → Ines is a solvilaw.
- Every solvilaw is a didreven. → Ines is a didreven.
- Every didreven is a jolvuxoj. → Ines is a jolvuxoj.
- Every jolvuxoj is a xotlukuk. → Ines is a xotlukuk.
- Every xotlukuk is a tojnawos. → Ines is a tojnawos.
- Every goskomab is a xankivut. → Ines is a xankivut.
- Every xankivut is a vizmotir. → Ines is a vizmotir.
- Every vizmotir is a lanzurup. → Ines is a lanzurup.
- Every lanzurup is a verkobip. → Ines is a verkobip.
- Every verkobip is a xorzezof. → Ines is a xorzezof.
- Every xorzezof is a mujnalas. → Ines is a mujnalas.
- Every tojnawos that is also a mujnalas is a xovkuvat. → Ines is a xovkuvat.
- Every goskomab is a xombokaz. → Ines is a xombokaz.
- Every xombokaz is a vivkabaz. → Ines is a vivkabaz.
- Every vivkabaz is a rizmizur. → Ines is a rizmizur.
- Every rizmizur is a wavkebif. → Ines is a wavkebif.
- Every wavkebif is a rizlabed. → Ines is a rizlabed.
- Every rizlabed is a solvilaw. → Ines is a solvilaw.
- Every solvilaw is a didreven. → Ines is a didreven.
- Every didreven is a jolvuxoj. → Ines is a jolvuxoj.
- Every jolvuxoj is a xotlukuk. → Ines is a xotlukuk.
- Every goskomab is a xankivut. → Ines is a xankivut.
- Every xankivut is a vizmotir. → Ines is a vizmotir.
- Every vizmotir is a lanzurup. → Ines is a lanzurup.
- Every lanzurup is a verkobip. → Ines is a verkobip.
- Every verkobip is a xorzezof. → Ines is a xorzezof.
- Every xorzezof is a mujnalas. → Ines is a mujnalas.
- Every mujnalas is a vuskimaf. → Ines is a vuskimaf.
- Every vuskimaf is a likribuj. → Ines is a likribuj.
- Every likribuj is a gurdekir. → Ines is a gurdekir.

- Every gurdekir is a nazlazal. → Ines is a nazlazal.
- Every xotlukuk that is also a nazlazal is a veskakot. → Ines is a veskakot.
- Every xovkuvat that is also a veskakot is a xejnuwur. → Ines is a xejnuwur.
- Every goskomab is a xankivut. → Ines is a xankivut.
- Every xankivut is a vizmotir. → Ines is a vizmotir.
- Every vizmotir is a lanzurup. → Ines is a lanzurup.
- Every lanzurup is a verkobip. → Ines is a verkobip.
- Every verkobip is a xorzezof. → Ines is a xorzezof.
- Every xorzezof is a mujnalas. → Ines is a mujnalas.
- Every mujnalas is a vuskimaf. → Ines is a vuskimaf.
- Every vuskimaf is a likribuj. → Ines is a likribuj.
- Every likribuj is a gurdekir. → Ines is a gurdekir.
- Every gurdekir is a nazlazal. → Ines is a nazlazal.
- Every nazlazal is a pugnokik. → Ines is a pugnokik.
- Every pugnokik is a membejoj. → Ines is a membejoj.
- Every membejoj is a kimbeluv. → Ines is a kimbeluv.
- Every xejnuwur that is also a kimbeluv is a warkepon. → Ines is a warkepon.
- Every goskomab is a zoltiwud. → Ines is a zoltiwud.
- Every zoltiwud is a bambemuk. → Ines is a bambemuk.
- Every bambemuk is a rerkazav. → Ines is a rerkazav.
- Every rerkazav is a rujnogug. → Ines is a rujnogug.
- Every rujnogug is a kugnotov. → Ines is a kugnotov.
- Every kugnotov is a vezlixid. → Ines is a vezlixid.
- Every vezlixid is a buxtagem. → Ines is a buxtagem.
- Every buxtagem is a sivkulos. → Ines is a sivkulos.
- Every sivkulos is a xembupeg. → Ines is a xembupeg.
- Every xembupeg is a pidrokis. → Ines is a pidrokis.
- Every pidrokis is a nizmozim. → Ines is a nizmozim.
- Every warkepon that is also a nizmozim is a ninkazij. → Ines is a ninkazij.

Key says: **Ines is a ninkazij** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 9 (problem P7). Start: *Dev is a jujnapot.*

- Every jujnapot is a mevkodot. → Dev is a mevkodot.
- Every mevkodot is a suzlepij. → Dev is a suzlepij.
- Every suzlepij is a kerkelem. → Dev is a kerkelem.
- Every kerkelem is a vodruzux. → Dev is a vodruzux.
- Every vodruzux is a markajif. → Dev is a markajif.
- Every markajif is a renzolox. → Dev is a renzolox.
- Every renzolox is a bavkared. → Dev is a bavkared.
- Every bavkared is a vivkamix. → Dev is a vivkamix.
- Every vivkamix is a derzinon. → Dev is a derzinon.

- Every derzinon is a jizlejob. → Dev is a jizlejob.
- Every jujnapot is a gixtavoz. → Dev is a gixtavoz.
- Every gixtavoz is a pijnupev. → Dev is a pijnupev.
- Every pijnupev is a perzijez. → Dev is a perzijez.
- Every perzijez is a negnoxek. → Dev is a negnoxek.
- Every negnoxek is a fozmegax. → Dev is a fozmegax.
- Every fozmegax is a damboxig. → Dev is a damboxig.
- Every jizlejob that is also a damboxig is a xirzujin. → Dev is a xirzujin.
- Every jujnapot is a lanzodol. → Dev is a lanzodol.
- Every lanzodol is a zilvizol. → Dev is a zilvizol.
- Every zilvizol is a sutlokob. → Dev is a sutlokob.
- Every sutlokob is a zadrulek. → Dev is a zadrulek.
- Every zadrulek is a gizmusuk. → Dev is a gizmusuk.
- Every gizmusuk is a mejnulor. → Dev is a mejnulor.
- Every mejnulor is a tiskelik. → Dev is a tiskelik.
- Every tiskelik is a favkuwev. → Dev is a favkuwev.
- Every favkuwev is a wagnafov. → Dev is a wagnafov.
- Every wagnafov is a kirdovet. → Dev is a kirdovet.
- Every kirdovet is a tujnajoz. → Dev is a tujnajoz.
- Every xirzujin that is also a tujnajoz is a nixtepun. → Dev is a nixtepun.

Key says: **Dev is a nixtepun** is entailed. ☐ correct ☐ wrong ☐ unclear

Item 10 (problem P6). Start: *Tobin is a rukraruz.*

- Every rukraruz is a malvodub. → Tobin is a malvodub.
- Every malvodub is a kazlewig. → Tobin is a kazlewig.
- Every kazlewig is a jozmafel. → Tobin is a jozmafel.
- Every jozmafel is a veltadon. → Tobin is a veltadon.
- Every veltadon is a pezlekir. → Tobin is a pezlekir.
- Every pezlekir is a finkitiw. → Tobin is a finkitiw.
- Every finkitiw is a korkigun. → Tobin is a korkigun.
- Every korkigun is a fotlexur. → Tobin is a fotlexur.
- Every rukraruz is a pombawid. → Tobin is a pombawid.
- Every pombawid is a zurzozir. → Tobin is a zurzozir.
- Every zurzozir is a nankexid. → Tobin is a nankexid.
- Every fotlexur that is also a nankexid is a sunzuwes. → Tobin is a sunzuwes.

Key says: **Tobin is a sunzuwes** is entailed. ☐ correct ☐ wrong ☐ unclear

Part 2. Is the fact NOT entailed?

In each item a join rule has only one of its two conditions true for the target. The key says the conclusion is not entailed.

Item 11 (problem P6). Rule: *Every seskobar that is also a kajnates is a lazlopon.* The target Tobin is a seskobar (derived), but nothing makes Tobin a kajnates.

Key says: **Tobin is a lazlopon** is NOT entailed. ☐ correct ☐ wrong ☐ unclear

Item 12 (problem P5). Rule: *Every bagnumox that is also a mozmubon is a lulvepum.* The target Ines is a mozmubon (derived), but nothing makes Ines a bagnumox.

Key says: **Ines is a lulvepum** is NOT entailed. ☐ correct ☐ wrong ☐ unclear

Item 13 (problem P1). Rule: *Every dignukaf that is also a zozlenok is a jignukol.* The target Orla is a dignukaf (derived), but nothing makes Orla a zozlenok.

Key says: **Orla is a jignukol** is NOT entailed. ☐ correct ☐ wrong ☐ unclear

Item 14 (problem P4). Rule: *Every borzavet that is also a kurzixuj is a xixtunos.* The target Nell is a kurzixuj (derived), but nothing makes Nell a borzavet.

Key says: **Nell is a xixtunos** is NOT entailed. ☐ correct ☐ wrong ☐ unclear

Result

Items correct: ___ / 14. Anything wrong or unclear (item numbers and why):
_____